Informatica® Data Ingestion and Replication
October 2026

Getting Started with Data Ingestion and Replication

Informatica Data Ingestion and Replication Getting Started with Data Ingestion and Replication
October 2026

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Preface

Read *Getting Started with Data Ingestion and Replication* for information about the steps you need to perform before you begin configuring data ingestion and replication tasks. It assumes that you have a working knowledge of IDMC.

Informatica Resources

Informatica provides you with a range of product resources through the Informatica Network and other online portals. Use the resources to get the most from your Informatica products and solutions and to learn from other Informatica users and subject matter experts.

Informatica Documentation

Use the Informatica Documentation Portal to explore an extensive library of documentation for current and recent product releases. To explore the Documentation Portal, visit https://docs.informatica.com.

If you have questions, comments, or ideas about the product documentation, contact the Informatica Documentation team at infa_documentation@informatica.com.

IDMC web site

You can access the Informatica® Intelligent Data Management Cloud™ (IDMC) web site at http://www.informatica.com/cloud.

IDMC Communities

Use the IDMC Community to discuss and resolve technical issues. You can also find technical tips, documentation updates, and answers to frequently asked questions.

Access the IDMC Community at:

https://network.informatica.com/community/informatica-network/products/cloud-integration

Developers can learn more and share tips at the Cloud Developer community:

https://network.informatica.com/community/informatica-network/products/cloud-integration/cloud-developers

IDMC Marketplace

Visit the Informatica Marketplace to try and buy Data Integration Connectors, templates, and mapplets:

https://marketplace.informatica.com/

Data Integration connector documentation

You can access documentation for Data Integration Connectors at the Documentation Portal. To explore the Documentation Portal, visit https://docs.informatica.com.

Informatica Knowledge Base

Use the Informatica Knowledge Base to find product resources such as how-to articles, best practices, video tutorials, and answers to frequently asked questions.

To search the Knowledge Base, visit https://search.informatica.com. If you have questions, comments, or ideas about the Knowledge Base, contact the Informatica Knowledge Base team at KB_Feedback@informatica.com.

IDMC Trust Center

The IDMC Trust Center provides information about Informatica security, compliance, and privacy practices.

You can access the trust center at https://trust.informatica.com/.

IDMC Pulse Dashboard

The IDMC Pulse Dashboard provides system status and upgrade information for all PODs and services across IDMC.

Follow the services you use and subscribe to alerts to receive notifications about upgrades, maintenance, production incidents, and upcoming releases. The IDMC Pulse Dashboard displays the production status of all the IDMC products. All maintenance updates are posted to this page, and during an outage, it will have the most current information. To ensure you are notified of updates and outages, you can subscribe to receive updates for a single service or all IDMC services. Subscribing to all services is the best way to be certain you never miss an update.

For more information, see the Service Status and Communications guide.

Informatica Global Customer Support

You can contact a Global Support Center through the Informatica Network or by telephone.

To find online support resources on the Informatica Network, click **Contact Support** in the IDMC Help menu to go to the **Cloud Support** page. The **Cloud Support** page includes system status information and community discussions. Log in to Informatica Network and click **Need Help** to find additional resources and to contact Informatica Global Customer Support through email.

The telephone numbers for Informatica Global Customer Support are available from the Informatica web site at https://www.informatica.com/services-and-training/support-services/contact-us.html.

CHAPTER 1

Getting Started with Data Ingestion and Replication

Before you configure a Data Ingestion and Replication task, verify that all prerequisite steps have been completed. Most of these steps are completed in the common Administrator service.

Step 1. Check system requirements

Check the following items:

- For Database Ingestion and Replication minimum system requirements, see *Database Ingestion and Replication > Database Ingestion and Replication system requirements*.
- To determine the web browsers that are compatible with Data Ingestion and Replication and the operating systems that are supported for the Secure Agent, check the Product Availability Matrix (PAM) for IDMC (IICS) on the Informatica Network at https://network.informatica.com/community/informatica-network/product-availability-matrices/
- To determine the source and target types and versions that are supported for each ingestion type, check the KB article What are the supported sources and targets for IICS Cloud Mass Ingestion service?.

Step 2. Set up an organization

If you are the administrator, set up an organization from the **Organization** page in Administrator. An organization is a secure area within the IDMC repository that stores your licenses, user accounts, ingestion tasks, and information about jobs and security.

Then, configure users, user groups, and user role permissions for the organization.

If your organization has the Organization Hierarchy license, you can also create one or more sub-organizations within your organization. You can create sub-organizations to represent different business environments in your company. For example, you might create separate sub-organizations to represent your development, testing, and production environments.

For more information, see "Organizations" in the Administrator help.

Step 3. Download and install a Secure Agent

On the **Runtime Environments** page in Administrator, download a local Secure Agent and install it. A Secure Agent is a lightweight program that runs tasks and enables secure communication across the firewall between your organization and IDMC.

You can use a single Secure Agent or a Secure Agent group with multiple agents. When you download and install a Secure Agent, a *Secure Agent group*, also called a runtime environment, is created. A Secure Agent group can contain one Secure Agent or multiple agents if your licensing allows it.

To achieve the best performance in high-volume or performance-critical environments, run Data Ingestion and Replication tasks on a Secure Agent or Secure Agent group that's dedicated to Data Ingestion and Replication, that is, not shared with any other products or processes.

After you enable services and connectors for the Secure Agent group, the components and packages based on your selections are downloaded and an associated runtime environment is created. For more information, see "Secure Agent Groups" in the Administrator help.

If you use a Secure Agent group with multiple agents, an available agent will be picked from the active agent list to run your jobs. The use of multiple agents is supported for application and database ingestion and replication initial load jobs with any source type and for database ingestion and replication incremental load and combined initial and incremental load jobs with a Db2 for i, Db2 for z/OS, Oracle, or SQL Server source subject to some limitations. For information about automatic agent switchovers, see "Agent failover in a Secure Agent group" on page 31.

If you are an administrator of a parent organization, you can share the parent organization's Secure Agent group with sub-organizations. However, a sub-organization user cannot share a Secure Agent group. When a Secure Agent group is shared, users in the sub-organizations can create application or database ingestion and replication tasks of any load type and run the associated jobs on the Secure Agents in the group. Sharing a Secure Agent group can help optimize the use of available Secure Agent resources. For more information, see "Shared Secure Agent groups" in the Administrator help.

Guidelines:

- If you remove a Secure Agent from a Secure Agent group and add it to another Secure Agent group, wait at least 60 minutes before trying to use it to run application or database ingestion and replication jobs. Application Ingestion and Replication and Database Ingestion and Replication refresh the internal cache of Secure Agent runtime information every hour by default.
- If you use Secure Agent sub-organizations, you cannot unlink a sub-organization from a parent organization and then relink it to another parent organization. If you do, problems such as duplicate run IDs can occur for your jobs.
- If you create a connection using a shared Secure Agent group as the runtime environment and create an application or database ingestion and replication task with that connection under a sub-organization that uses a non-shared Secure Agent group, the task wizard can't fetch the list of schemas for the target. The connection and task must both use a shared Secure Agent group, either the same shared group or different shared groups.

Step 4. Configure the runtime environment

On the **Runtime Environments** page in Administrator, create a new runtime environment, or manage existing runtime environments. The **New Runtime Environment** option lets you create a Secure Agent group or a Customer Managed Secure Agent group. Use the **Serverless** tab to configure a serverless runtime environment.

A runtime environment is the execution platform for running tasks. You must have at least one runtime environment in your organization for users to be able to run tasks. Make sure the runtime environment you want to use is running. If you create another runtime environment, you must add an unassigned Secure Agent to it.

Data Ingestion and Replication can use the following runtime environments:

- **Secure Agent Group.** Use this runtime environment for connections and tasks in Database Ingestion and Replication. A Secure Agent within the group runs the tasks.
- **Customer Managed Runtime Group.** Use this runtime environment for application ingestion and replication tasks, database ingestion and replication tasks, or file ingestion and replication tasks. Secure agents on this runtime environment run with no dependency on a connection to IDMC.

  Note: You can't use a Customer Managed Runtime Group for streaming ingestion and replication tasks.

- **Serverless runtime environment.** If your cloud environment is Azure, you can configure a serverless runtime environment for application ingestion and replication tasks and database ingestion and replication tasks that have a supported source and target type. This environment is hosted by Informatica, so you don't need to configure a Secure Agent or Secure Agent group. For more information, see "Serverless runtime environments" on page 32 and Data Ingestion and Replication *Connectors and Connections* documentation for your connector.

  Note: Data Ingestion and Replication can't use the Informatica Cloud Hosted Agent.

  For more information about runtime environments, see *Administrator* > *Runtime environments*.

Under **System Configuration Details**, configure properties for the CMI Streaming Agent, Database Ingestion, or Mass Ingestion (file ingestion) service. For more information, see "Secure Agent services" on page 16.

Step 5. Configure connections

From the **Connections** page in Administrator, configure connection properties for the source and target connectors that you want to use in Data Ingestion and Replication tasks. To quickly get to the **New Connection** page, you can click the **Let's get connected!** pane on the Home page. For more information, see *Connectors and Connections* > *Data Ingestion and Replication connectors* and *Data Ingestion and Replication connection properties*.

Step 6. Create your project

On the **Explore** page , you can create projects and project folders to organize your Data Ingestion and Replication tasks. To access the Explore page, click **Explore** in the left navigation bar of the Home page. A project can contain multiple subfolders. See "Creating projects and project folders" on page 34.

Installing Secure Agents

You can install Secure Agents on Windows or Linux.

Installing on Windows

On Windows, the Secure Agent runs as a Windows service. When you install the Secure Agent, you also install the Informatica Cloud Secure Agent Manager.

By default, the Secure Agent starts when you start Windows. You can stop and restart the Secure Agent using the Secure Agent Manager or Windows Services. If you install the Secure Agent on a different volume than you use to run the installation program, you must start and stop the Secure Agent from Windows Services.

You can also use the Secure Agent Manager to check the Secure Agent status and configure proxy information. The Secure Agent works with BASIC, DIGEST, and NTLMv2 proxy authentication.

You can launch the Secure Agent Manager from the Start menu or desktop icon. When you close the Secure Agent Manager, it minimizes to the Windows taskbar notification area for quick access.

When you install a Secure Agent, you perform the following tasks:

1. Verify that the machine meets the minimum requirements.
2. Download the Secure Agent installer files.
3. Install and register the Secure Agent.

Secure Agent requirements on Windows

You can install the Secure Agent on any machine that has internet connectivity and can access IDMC.

Verify the following requirements before you install the Secure Agent on Windows:

- The Secure Agent machine uses a supported operating system. For the list of supported operating systems for the Secure Agent, see the Product Availability Matrix (PAM) for Intelligent Data Management Cloud on the Knowledge Base.
- The Secure Agent machine has the following attributes at a minimum:
  - x86 64-bit architecture
  - 4 CPU cores
  - 16 GB RAM
  - 5 GB free disk space
- The Secure Agent machine is on a volume with at least 250GB disk space, with at least 5 GB free space or three times the size of the Secure Agent installation, whichever is greater.
- The account you use to install the Secure Agent has access to all remote directories that contain flat source or target files.
- No other Secure Agent is installed on the machine. If another Secure Agent is installed on the machine, uninstall it first.

For more information, see this KB article:
Minimum requirements and best practices when installing Informatica Cloud Secure Agent.

Configure the firewall

If your organization uses a protective firewall, include the IDMC domain name or IP address ranges in the list of approved domain names or IP addresses. To ensure that the Secure Agent can perform all necessary tasks through the firewall, enable the port that the Secure Agent uses.

The Secure Agent uses port 443 (HTTPS) to connect to the internet. Configure your firewall to allow traffic to pass over port 443. Also exclude the following directory from security scans: `<Secure Agent installation directory>/apps`. Scanning files in this directory can interfere with the operation of the Secure Agent.

The list of allowed domains and IP addresses can vary according to your POD (Point of Deployment). You can identify your POD through the URL that appears when you open any service. The first few characters of the URL string identify the POD. For example, if the URL starts with `usw3.dm-us.informaticacloud.com`, your POD is USW3.

For the list of all the domains and IP addresses that you must allow for the different PODs, see Pod Availability and Networking.

Setting Secure Agent permissions on Windows

A Secure Agent requires certain permissions to transfer data between sources and targets.

When you install a Secure Agent on Windows, the Secure Agent must be part of the local Administrators group.

Configuring Windows settings

Before you use the Secure Agent on Windows, configure proxy settings and a Windows Secure Agent service login.

You can configure proxy settings in Secure Agent Manager. Configure a login for the Windows Secure Agent service on Windows.

Note: If you use the Secure Agent for Informatica Cloud Data Wizard, you do not need to configure proxy settings or a Windows service login for the Secure Agent.

Downloading and installing the Secure Agent on Windows

To install the Secure Agent on a Windows machine, download and run the Secure Agent installer and then register the agent.

Before you download and install the Secure Agent, verify that no other Secure Agent is installed on the machine. If any other Secure Agent exists, you must uninstall it first.

Tip: To verify the checksum of the Secure Agent installation program, use the agent REST API version 2 resource. For more information about the agent resource, see *REST API Reference*.

1. Open Administrator and select **Runtime Environments**.
2. On the **Runtime Environments** page, click **Download Runtime Installer**.
3. Select **Secure Agent** as the environment type.
4. Select **Windows 64** as the platform.
5. Click **Copy** to copy the install token.

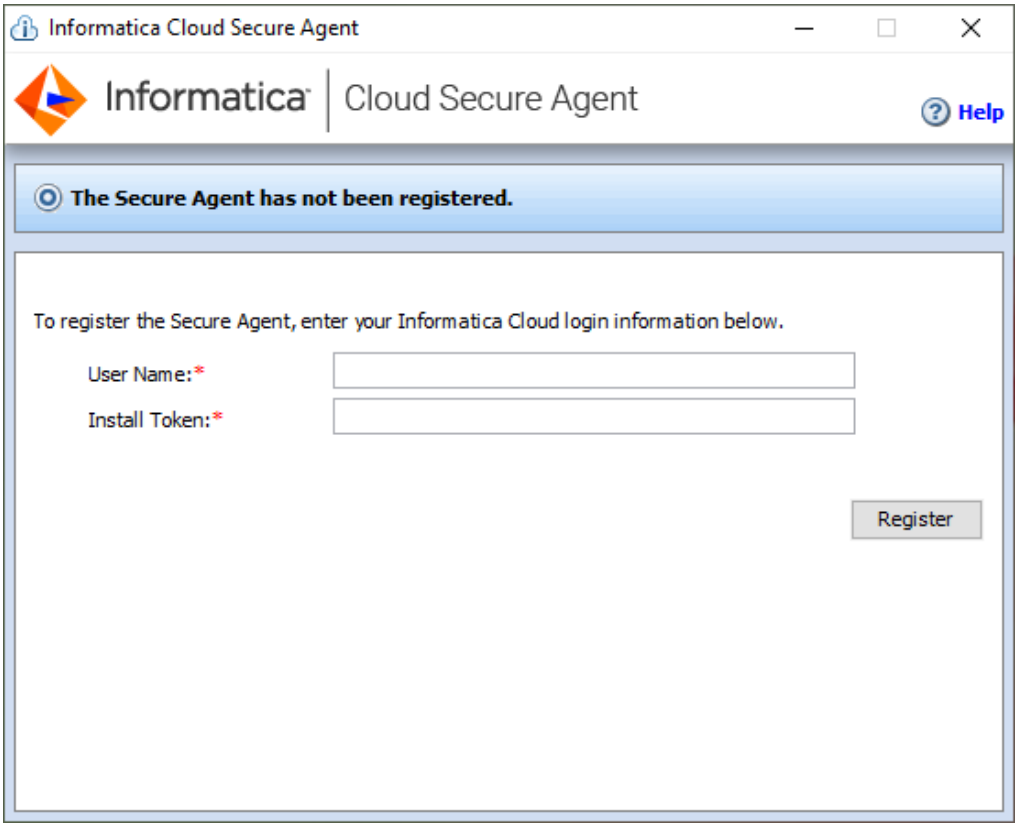
   You will need this token to install the Secure Agent. The token expires after 24 hours. If the token expires before you install the Secure Agent, repeat steps 1 to 5 to obtain a new token. You don't need to download the installer again.
6. Click **Download**.

   The installer is downloaded to your machine. The name of the installer is `agent64_install_ng_ext.<agent core version>.exe`.
7. Run the installer as an Administrator and complete the following steps:
   a. Specify the Secure Agent installation directory, and click **Next**.

      Note: Ensure that the file path doesn't contain multibyte characters. If the path contains multibyte characters, the Secure Agent might not start.

   b. Click **Install** to install the agent.

The **Cloud Secure Agent** dialog box opens and prompts you to register the agent as shown in the following image:



8. In the Secure Agent Manager, enter the following information, and then click **Register**:

Option	Description
User Name	User name that you use to access IDMC.
Install Token	Token that you copied when you downloaded the runtime installer.

The Secure Agent Manager displays the status of the Secure Agent. It takes a minute for all of the services to start.

Tip: When you register the agent, it is added to its own Secure Agent group by default. You can add the agent to a different Secure Agent group if necessary.

9. If your organization uses an outgoing proxy server to connect to the internet, enter the proxy server information.
10. Close the Secure Agent Manager.

    The Secure Agent Manager minimizes to the taskbar and continues to run as a service until stopped.

Updating proxy settings through the Secure Agent Manager

To update proxy hosts, proxy ports, and the username and password used to connect to the proxy server, edit the proxy settings through the Secure Agent Manager.

1. In the Secure Agent Manager, click **Proxy**.
2. To enter proxy server settings, click **Use a Proxy Server**.
3. Enter the following information:

Field	Description
Proxy Host	Required. Host name of the outgoing proxy server that the Secure Agent uses.
Proxy Port	Required. Port number of the outgoing proxy server.
User Name	User name to connect to the outgoing proxy server.
Password	Password to connect to the outgoing proxy server.

4. Click **OK**.

   The Secure Agent Manager restarts the Secure Agent to apply the settings.

Configuring a login for a Windows Secure Agent service

On Windows, configure a network login for the Secure Agent service. The Secure Agent can access the network with the privileges and permissions associated with the login.

Configure a login for the Secure Agent machine to allow the Secure Agent to access directories to configure and run tasks. When you configure connections, configure tasks, and run tasks that use flat file or FTP/SFTP connection types, the Secure Agent might require read and write permissions on the related directories.

For example, to browse to a directory to configure a flat file or FTP/SFTP connection, the Secure Agent login might require permission to access the directory. Without a Secure Agent login with the appropriate permissions, IDMC cannot display the directory in the **Browse for Directory** dialog box.

1. Go to the **Services** window from the Windows Administrative tools.
2. In the **Services** window, right-click the Informatica Cloud Secure Agent service and choose **Properties**.
3. In the **Properties** dialog box, click the **Log On** tab.
4. To configure a login, select **This Account**.
5. Enter an account and password.

   Use an account with the required privileges and permissions for the network security defined for the domain. By default, the account format is <domain name>\<user name>.
6. Click **OK**.
7. In the **Services** window, restart the Secure Agent service for the changes to take effect.

Uninstalling the Secure Agent on Windows

You can uninstall the Secure Agent. You might uninstall the Secure Agent if you no longer want to run the Secure Agent on the machine or if you want to reinstall the Secure Agent.

Before you uninstall the Secure Agent, verify that no connection or task is configured to use it.

1. Click **Start** > **All Programs** > **Informatica Cloud Secure Agent** > **Uninstall Informatica Cloud Secure Agent**.

   The Secure Agent uninstaller launches.
2. Click **Uninstall**.
3. When the uninstall completes, click **Done**.
4. Delete any remaining files in the installation directory.

   After you uninstall the Secure Agent, delete all files and directories associated with the Secure Agent installation.

Note: Uninstalling the Secure Agent does not delete log files from the Secure Agent directory. If you want to reinstall a Secure Agent on the machine, you must delete all files and directories associated with the Secure Agent installation or reinstallation will fail. If you want to save the log files, copy them to a different directory, and then delete the Secure Agent installation directory.

Installing on Linux

On Linux, the Secure Agent runs as a process. You can use a shell command line to install, register, start, stop, and uninstall the Secure Agent.

You can also use the shell command line to check the Secure Agent status.

When you install a Secure Agent, you perform the following tasks:

1. Verify that the machine meets the minimum requirements.
2. Download the Secure Agent installer files.
3. Install and register the Secure Agent.

Consider the following guidelines:

- Create a specific user profile to install the Secure Agent with full access to all folders from the Secure Agent installation directory. Don't install the Secure Agent as the root user.
- You can't install more than one Secure Agent on the same machine under the same user account. Multiple agents may exist under different user accounts.
- Don't install the Secure Agent on any node within the Informatica domain.

Secure Agent requirements on Linux

You can install the Secure Agent on any machine that has internet connectivity and can access IDMC. Before you install the Secure Agent on Linux, verify the system requirements.

Verify the following requirements before you install the Secure Agent on Linux:

- Verify that the Secure Agent machine uses a supported operating system. For the list of supported operating systems for the Secure Agent, see the Product Availability Matrix (PAM) for Intelligent Data Management Cloud on the Knowledge Base.
- Verify that the Secure Agent machine is running the x86 64-bit architecture, with at least 11 GB free disk space.

- Verify that the `libidn.x86_64` package is installed.
  If the package isn't present, install it using the following command: `sudo yum install libidn.x86_64`

  Note: The command to install the package might vary based on your Linux distribution.
- Verify that the `libidn.so.*` libraries are installed.
  If the libraries aren't present, run the following commands:
  1. Change to the appropriate directory on the Secure Agent machine.
     - For 64-bit systems: `cd /usr/lib/x86_64-linux-gnu`
     - For 32-bit systems: `cd /usr/lib/i386-linux-gnu`
  2. Create a symbolic link using the following command:

     ```
     sudo ln -s libidn.so.12 libidn.so.11
     ```

  If you are installing the Secure Agent on RHEL 9, create an additional symbolic link using the following command:

  ```
  sudo ln -s libidn2.so.0 libidn.so.11
  ```
- The account that you use to install the Secure Agent must have access to all remote directories that contain flat source or target files.
- If you use PowerCenter, install the Secure Agent using a different user account than the account you used to install PowerCenter.
  IDMC and PowerCenter use some common environment variables. If the environment variables are not set correctly for IDMC, your jobs might fail at run time.

For more information, see this KB article:
Minimum requirements and best practices when installing Informatica Cloud Secure Agent.

Configure the firewall

If your organization uses a protective firewall, include the IDMC domain name or IP address ranges in the list of approved domain names or IP addresses. To ensure that the Secure Agent can perform all necessary tasks through the firewall, enable the port that the Secure Agent uses.

The Secure Agent uses port 443 (HTTPS) to connect to the internet. Configure your firewall to allow traffic to pass over port 443. Also exclude the following directory from security scans: `<Secure Agent installation directory>/apps`. Scanning files in this directory can interfere with the operation of the Secure Agent.

The list of allowed domains and IP addresses can vary according to your POD (Point of Deployment). You can identify your POD through the URL that appears when you open any service. The first few characters of the URL string identify the POD. For example, if the URL starts with `usw3.dm-us.informaticacloud.com`, your POD is USW3.

For the list of all the domains and IP addresses that you must allow for the different PODs, see Pod Availability and Networking.

Setting Secure Agent permissions on Linux

A Secure Agent requires certain permissions to transfer data between sources and targets.

When you install a Secure Agent on Linux, the Secure Agent must have read/write/execute permissions for the installation directory.

Downloading and installing the Secure Agent on Linux

To install the Secure Agent on a Linux machine, download and run the Secure Agent installer and then register the agent.

Before you download and install the Secure Agent, verify that no other Secure Agent is installed on the machine using the same Linux user account. If any other Secure Agent exists, you must uninstall it first.

Tip: To verify the checksum of the Secure Agent installation program, use the agent REST API version 2 resource. For more information about the agent resource, see *REST API Reference*.

1. Open Administrator and select **Runtime Environments**.
2. On the **Runtime Environments** page, click **Download Runtime Installer**.
3. Select **Secure Agent** as the Environment Type.
4. Select **Linux 64** as the Platform.
5. Click **Copy** to copy the install token.

   You will need this token to install the Secure Agent. The token expires after 24 hours. If the token expires before you install the Secure Agent, repeat steps 1 to 5 to obtain a new token. You don't need to download the installer again.
6. Click **Download**.

   The installer is downloaded to your machine. The name of the installer is `agent64_install_ng_ext.<agent core version>.bin`.
7. Save the installer to a directory on the machine where you want to run the Secure Agent.

   Note: Ensure that the file path doesn't contain spaces or multibyte characters. If the file path contains spaces, the installation might fail. If the path contains multibyte characters, the Secure Agent might not start.
8. From a shell command line, navigate to the directory where you downloaded the installation program and enter the following command:

   ```
   ./agent64_install_ng_ext.bin -i console
   ```
9. When the installer completes, navigate to the following directory:

   ```
   <Secure Agent installation directory>/apps/agentcore
   ```
10. To start the Secure Agent, enter the following command:

    ```
    ./infaagent.sh startup
    ```

    The Secure Agent Manager starts. Register the agent using the user name that you use to access IDMC and also supply the install token.
11. To register the agent, in the `<Secure Agent installation directory>/apps/agentcore` directory, enter one of the following commands using your IDMC user name and the token that you copied:

    - To add the agent to its own Secure Agent group, use the following command:

      ```
      ./consoleAgentManager.sh configureToken <user name> <install token>
      ```
    - To add the agent to an existing Secure Agent group, use the following command:

      ```
      ./consoleAgentManager.sh configureTokenWithRuntime <user name> <install token>
      <Secure Agent group name>
      ```

      Note: If the command includes a Secure Agent group name that doesn't exist, the Secure Agent is not assigned to a group. Be sure to use a valid Secure Agent group name.

The following table lists the command options:

Option	Description
User Name	Required. IDMC user name of the user installing the Secure Agent.
Install Token	Required. The install token that you copied when you downloaded the runtime installer.
Secure Agent group name	Optional. Include a group name when you want to add the agent to an existing Secure Agent group instead. If this option isn't included in the command, the agent will be in its own Secure Agent group.

You can check the registration status of a Secure Agent using the following command:

```
./consoleAgentManager.sh isConfigured
```

Updating proxy settings through the command line

To update proxy hosts, proxy ports, and the username and password to connect to the proxy server, use the command line to update the proxy configuration file.

1. Navigate to the following directory:

   ```
   <Secure Agent installation directory>/apps/agentcore/conf
   ```

2. To update the proxy configuration file, `proxy.ini`, run the following command:

   ```
   ./consoleAgentManager.sh configureProxy <proxy host> <proxy port> <proxy user name>
   <proxy password>
   ```

3. Restart the Secure Agent.

Uninstalling the Secure Agent on Linux

You can uninstall the Secure Agent. You might uninstall the Secure Agent if you no longer want to run the Secure Agent on the machine or if you want to reinstall the Secure Agent.

Before you uninstall the Secure Agent, verify that no connection or task is configured to use it.

1. From the command line, navigate to the following directory:

   ```
   <Secure Agent installation directory>/apps/agentcore
   ```

2. Stop the Secure Agent Linux process by entering the following command:

   ```
   ./infaagent.sh shutdown
   ```

3. To uninstall the Secure Agent, run rm -rf on the directory where you installed the Secure Agent to remove Secure Agent files.

Secure Agent services

Secure Agent services are pluggable microservices that the Secure Agent uses for data processing. Each Secure Agent service runs independently of the other services that run on the agent.

The independent services architecture provides the following benefits:

- The Secure Agent does not restart when you add a connector or package.
- Services are not impacted when another service restarts.

- Downtime during upgrades is minimized. The upgrade process installs a new version of the Secure Agent, updates connector packages, and applies configuration changes for the services. To minimize downtime, the old agent remains available and continues to run ingestion jobs during the upgrade. The new version of the Secure Agent runs jobs that start after the upgrade process completes.

The services that run on a Secure Agent vary based on your licenses and the IDMC that your organization uses. For Data Ingestion and Replication, the following Secure Agent services are available:

- Database Ingestion - For running application ingestion and replication jobs and database ingestion and replication jobs
- CMI Streaming Agent - For running streaming ingestion and replication jobs
- Mass Ingestion - For running file ingestion and replication jobs

Each service has a unique set of configuration properties. You might need to configure a service or change the service properties to optimize performance or if you are instructed to do so by Informatica Global Customer Support.

Database Ingestion service

Both Application Ingestion and Replication and Database Ingestion and Replication use the Database Ingestion agent service to run jobs.

After you download the Secure Agent to your runtime environment and enable the Database Ingestion service, the Database Ingestion packages are pushed to the on-premises system where the Secure Agent runs. You can then optionally configure properties for the Database Ingestion service that runs on the Secure Agent.

Database Ingestion service properties

To change or optimize the behavior of the Database Ingestion service that your Secure Agent group uses, you can configure Database Ingestion agent configuration properties for your runtime environment.

To configure the properties, open a Secure Agent in your runtime environment and click **Edit**. Under **System Configuration Details**, select **Database Ingestion** as the service and **DBMI_AGENT_CONFIG** or **LOCAL_TASK_CONFIG** as the type.

The following table describes the Database Ingestion agent service properties for the **DBMI_AGENT_CONFIG** type:

Property	Description
maxTaskUnits	The maximum number of application ingestion and replication task units and database ingestion and replication task units that can run concurrently on an on-premises machine where the Secure Agent is running. Task units are not related to the capacity and availability of your hardware or software. You can configure maxTaskUnits to precisely control CPU usage. Valid values are 1 to 2000000000 (2 billion). To calculate a reasonable number of task units for your Secure Agent machine, Informatica recommends that you divide the number of cores by 3 or 4. For example, if you have an 8-core machine, you could set this property to 2. Then monitor CPU usage and adjust the property value as needed to tune performance. During initial load processing, this property determines the number of tables that can be unloaded simultaneously. Remaining tables are queued and start unload processing when resources become available. **Note:** A single job can process many tables. The total number of tables that can be processed is limited only by available memory. On the average, 25 MB of RAM is required per table for an initial load task based on a 1 KB row size. During incremental load processing, this property determines the number of application ingestion and replication and database ingestion and replication jobs that can run simultaneously. Setting this property to a value greater than the number of cores on the Secure Agent machine can increase parallelism for task execution but also cause performance bottlenecks at task execution time.
serviceLogRetentionPeriod	The number of days to retain each internal Database Ingestion service log file after the last update is written to the file. When this retention period elapses, the log file is deleted. The default value is 7 days. Service logs are retained on the Secure Agent host where they are created: `<infaagent>/apps/Database_Ingestion/logs`. **Note:** This property is applicable to both Application Ingestion and Replication and Database Ingestion and Replication.
taskLogRetentionPeriod	The number of days to retain each job log file after the last update is written to the file. When this retention period elapses, the log file is deleted. The default value is 7 days.
ociPath	For Oracle sources and targets, the path to the Oracle Call Interface (OCI) directory that contains the oci.dll or libclntsh.so file. By default, Oracle uses $ORACLE_HOME/lib on Linux or %ORACLE_HOME%\bin on Windows. The OCI library is used by database ingestion CDC tasks to connect to Oracle. For a DBMI agent that is running, this value is appended to the PATH environment variable value on Windows or to the LD_LIBRARY_PATH environment variable value on Linux. This property is not required if you already included the OCI path in the PATH or LD_LIBRARY_PATH environment variable. **Note:** This property is applicable only to Database Ingestion and Replication.
serviceUrl	The URL that the Database Ingestion service uses to connect to the IDMC cloud. **Note:** This property is applicable to both Application Ingestion and Replication and Database Ingestion and Replication.

Property	Description
logLevel	The level of detail to include in the logs that the Database Ingestion service produces. Options are: - TRACE - DEBUG - INFO - WARN - ERROR The default value is TRACE. **Note:** This property is applicable to both Application Ingestion and Replication and Database Ingestion and Replication.
taskExecutionHeapSize	The maximum heap size, in gigabytes, for the Task Execution service. This heap size is used for the single Java process that hosts all application ingestion and replication tasks and database ingestion and replication tasks that run in the common Container service. You might need to increase this value if you run a large number of tasks or process a large volume of data. Enter the property value followed by "g" for gigabytes. The default value is '8g'. **Note:** This property is applicable to both Application Ingestion and Replication and Database Ingestion and Replication.
useProxy	Set this property to true to force the Database Ingestion agent service (DBMI Agent) to use the proxy settings from the Secure Agent proxy configuration when connecting to sources and targets. When set to false, proxy information might still be used. The DBMI Agent allows the JDBC drivers to fall back to their own default proxy usage behavior, which can be triggered by other proxy-related environment variable settings. Default is false. **Note:** This property is applicable to both Application Ingestion and Replication and Database Ingestion and Replication.
intermediateStorageDirectory	For incremental load and combined initial and incremental load jobs, the local root directory under which intermediate files that contain data are stored when the **Enable Persistent Storage** option is selected in the associated task definitions. This directory must reside on a safe, highly accurate, and high-performance device, as the stored data is critical to the product success and operational accuracy. **Note:** This property is applicable only to Database Ingestion and Replication.
storageBackupDirectory	For incremental load and combined initial and incremental load jobs, the path to the directory that stores backup files when the **Enable Persistent Storage** option is selected in the associated task definitions. This directory must reside on a safe, highly accurate, and high-performance device, as the stored data is critical to the product success and operational accuracy. **Note:** This property is applicable only to Database Ingestion and Replication.

Property	Description
storageProperties	For incremental load and combined initial and incremental load jobs, a comma-separated list of key=value pairs that is used when the **Enable Persistent Storage** option is selected in the associated task definitions. Specify this property only at the direction of Informatica Global Customer Support. **Note:** This property is applicable only to Database Ingestion and Replication.
supportedLoadTypes	For application ingestion and replication jobs and database ingestion and replication jobs, the load types that the Database Ingestion agent service can process. You can enter one or more of the following values, separated by a comma (,): - INITIAL. Initial load jobs or the initial load phase of combined initial and incremental load jobs. - INCREMENTAL. Incremental load jobs or the incremental phase of combined initial and incremental load jobs, which write to your target. - INCREMENTAL_STAGING. CDC staging tasks of incremental load or combined load jobs. Default is INITIAL,INCREMENTAL,INCREMENTAL_STAGING, which indicates all load types. **Notes:** - If multiple Database Ingestion agent services are configured to support the same load types, the jobs use the agent with the most available task units. - If the Database Ingestion agent service where a combined load job is running does not support the initial load type, the initial load phase of the combined load job is directed to another agent in the Secure Agent group where the initial load type enabled.

The following table describes the Database Ingestion agent service properties for the **LOCAL_TASK_CONFIG** type:

Property	Description
taskExecutionHeapSize	The maximum heap size, in gigabytes, available for running a single task, such as the CDC Staging Task, in its own Java Virtual Memory (JVM). You might need to increase this size if the task processes a large volume of data. Enter the property value followed by "g" for gigabytes. The default value is '2g'.
taskStartTimeoutSeconds	The number of seconds that must elapse before an attempt to start a task in its own JVM times out. The default value is 120.

Database Ingestion Agent environment variables

To change or optimize the behavior of the Database Ingestion agent service, you can define environment variables:

To configure environment variables, open a Secure Agent in your runtime environment and click **Edit**. Under **System Configuration Details** or **Custom Configuration Details**, select **Database Ingestion** as the service and **DBMI_AGENT_ENV** as the type.

Environment Variable	Description
DBMI_REPLACE_UNSUPPORTED_CHARS	For Microsoft Azure Synapse Analytics targets, controls whether an application ingestion and replication job or database ingestion and replication job replaces characters in character data that the target cannot process correctly. To enable character replacement, set this environment variable to true. `DBMI_REPLACE_UNSUPPORTED_CHARS=true` Application Ingestion and Replication or Database Ingestion and Replication then uses the character that is specified in the DBMI_UNSUPPORTED_CHARS_REPLACEMENT environment variable to replace unsupported characters.
DBMI_UNSUPPORTED_CHARS_REPLACEMENT	If the DBMI_REPLACE_UNSUPPORTED_CHARS environment variable is set to true, specifies the character that replaces the characters in source data that a Microsoft Azure Synapse Analytics target cannot process correctly. Default value: ? (question mark) **Note:** Define this environment variable only for Database Ingestion and Replication.
DBMI_WRITER_CONN_POOL_SIZE	Indicates the number of connections that an application ingestion and replication job or database ingestion and replication job uses to propagate the change data to the target. The default value is 8. Valid values are 4 through 8.
DBMI_WRITER_RETRIES_MAX_COUNT	If a network issue occurs while a database ingestion and replication job is loading source data to an Amazon S3 or Microsoft Azure Data Lake Storage Gen2 target, indicates the maximum number of times that the job retries a request to continue the initial load or incremental load. If all of the retries fail, the job fails. The default value is 5.
DBMI_WRITER_RETRIES_INTERVAL_IN_MILLIS	Specifies the time interval, in milliseconds, that a database ingestion and replication job waits before retrying the request to continue the initial load or incremental load to an Amazon S3 or Microsoft Azure Data Lake Storage Gen2 target if a network issue occurs. The default value is 1000.

Note: After you define or change an environment variable, restart the Database Ingestion Agent for the changes to take effect.

Database Ingestion agent service availability during an upgrade

Two versions of the Database Ingestion service (DBMI Agent) can operate simultaneously under the Secure Agent to minimize downtime of application ingestion and replication jobs and database ingestion and replication jobs during an upgrade.

During startup, the newer DBMI Agent version initiates a shutdown of the older DBMI Agent version. During the shutdown,application and database ingestion and replication jobs that use the incremental load or combined load type are migrated to the newer DBMI Agent version, whereas initial load jobs are allowed to complete on the older DBMI Agent version. The older DBMI Agent version shuts down after all of the initial load jobs complete or after 24 hours. In summary:

- Incremental load and combined load jobs are stopped and restarted on the newer DBMI Agent version from where they last left off when using the older DBMI Agent version.
- Initial load jobs are allowed to run on the older DBMI Agent version until completion or up to the maximum of 24 hours. If the maximum time limit is met, the older DBMI Agent version is forcefully stopped and the jobs are migrated to the newer DBMI Agent version.

Because the newer DBMI Agent is available to accept ingestion and replication jobs during the upgrade process, the jobs experience minimal downtime.

If the newer version of the DBMI Agent fails to start, for example, because of low memory on the Secure Agent machine, the older version of the DBMI Agent continues running for up to 24 hours, ensuring that no service disruption occurs while startup of the light-weight DBMI Agent is retried.

Note that Database Ingestion agent service log files, including Metadata Manager,Task Container, and DBMI Agent log files, include the application package version to differentiate the logs generated by different versions.

Mass Ingestion (Files)

To change or optimize the behavior of File Ingestion and Replication that your Secure Agent group uses, configure Mass Ingestion agent service properties for your runtime environment in Administrator.

You can configure the following properties:

Type	Name	Description
AGENT_RUNTIME_SETTINGS	file-listener-snapshot-dir	A directory where the snapshots of a new file listener components are added. You can add the following directory paths: - A path relative to the `MassIngestionRuntime` directory. For example, `../data/monitor`. - The absolute path. For example, `<Secure agent installation directory>/apps/MassIngestionRuntime/data/monitor` where *Secure agent installation directory* is the name of the directory where the secure agent is installed. **Note:** Use the snapshot directory shared with all agents when multiple Secure Agents are present in a group.
AGENT_RUNTIME_SETTINGS	mi-task-workspace-dir	A directory in the agent that file ingestion and replication tasks use as an intermediate staging area when transferring files to a target. The directory is a custom location in the agent. The path can be a shared location, mounted location, or a location apart from the default location in the agent.

Type	Name	Description
AGENT_RUNTIME_SETTINGS	mi-task-project-dir	A directory where the file ingestion and replication task stores the project files. The directory is a custom location in the agent. The path can be a shared location, mounted location, or a location apart from the default location in the agent.
AGENT_RUNTIME_SETTINGS	mi-task-logs-dir	A directory where the file ingestion and replication task stores the task logs files. The directory is a custom location in the agent. The path can be a shared location, mounted location, or a location apart from the default location in the agent.
AGENT_RUNTIME_SETTINGS	mi-task-quarantine-dir	A directory where the file ingestion and replication task stores the infected files detected when you run a virus scan. The directory is a custom location in the agent. The path can be a shared location, mounted location, or a location apart from the default location in the agent. For example, userdata\quarantine **Note:** To automatically clean up the quarantine directory, set the agent property for the quarantine location to a system temporary files location such as `/tmp/informatica/fmi/quarantine`.
AGENT_RUNTIME_SETTINGS	agent-dedup-repository	The information about skipped duplicate files is saved in IDMC. To save the skipped duplicate files information in the Secure Agent, set the property to `true`. Default is `false`. For more information about saving the skipped duplicate information, see the *File Ingestion and Replication* guide.
AGENT_RUNTIME_SETTINGS	mi-dedup-snapshot-dir	Enter the path to store the information about skipped duplicate files in the Secure Agent. Applies only when the **agent-dedup-repository** property is set to `true`.
AGENT_RUNTIME_SETTINGS	file-listener-max-pool-size	The maximum number of threads to execute the file listener. Default is 20.
AGENT_RUNTIME_SETTINGS	file-listener-core-pool-size	The total number of threads. Default is 20.
AGENT_RUNTIME_SETTINGS	fmi-task-max-pool-size	The maximum number of threads to execute the file ingestion and replication task. Default is 50.
AGENT_RUNTIME_SETTINGS	fmi-task-core-pool-size	The initial or minimum number of threads. Default is 20.
AGENT_RUNTIME_SETTINGS	ftp-receive-socket-buffer-size	The buffer size for FTP inbound packets. Default is 16 bytes.

Type	Name	Description
AGENT_RUNTIME_SETTINGS	ftp-send-socket-buffer-size	The buffer size for FTP outbound packets. Default is 16 bytes.
AGENT_RUNTIME_SETTINGS	http-client-timeout	The timeout duration in seconds for Agent requests to IDMC. Default is 30 seconds.
PGP_SETTINGS	public-keyring-path	The directory to store the public key ring. You can add the following directory paths: - A path relative to the directory where Data Ingestion and Replication is installed. For example, `../data/pubring.pkr` where *pubring.pkr* is the name of the file where you store the public key ring. - The absolute path. For example, `<Secure agent installation directory>/apps/MassIngestionRuntime/data/pubring.pkr` where *pubring.pkr* is the name of the file where you store the public key ring and *Secure agent installation directory* is the name of the directory where the agent is installed.
PGP_SETTINGS	secret-keyring-path	The directory to store the secret key ring. You can add the following directory paths: - A path relative to the directory where Data Ingestion and Replication is installed. For example, `../data/secring.pkr` where *secring.pkr* is the name of the file where you store the secret key ring. - The absolute path. For example, `<Secure agent installation directory>/apps/MassIngestionRuntime/data/secring.pkr` where *secring.pkr* is the name of the file where you store the secret key ring and *Secure Agent installation directory* is the name of the directory where the agent is installed.
JVM_SETTINGS	app-heap-size	The minimum and maximum heap sizes of the File Ingestion and Replication application. Default is -Xms256m -Xmx2048m.
JVM_SETTINGS	lcm-heap-size	The minimum and maximum heap sizes of life-cycle management scripts. Default is -Xms32m -Xmx128m.

You can configure the following properties in the **Custom Configuration Details** area when you edit a Secure Agent:

Type	Name	Description
AGENT_RUNTIME_SETTINGS	ComplexFileDisableWriteChecksum	Set the value to **True** to ignore the `crc` file. The job runs successfully with Hadoop Files V2 as source and Snowflake Cloud Data Warehouse V2 as the target.

Guidelines to specify the folder path

A folder path can be a shared location, mounted location, or a location apart from the default location in the Secure Agent.

The following table lists the use of slashes around the source folder path:

Source	Folder Path
Windows	<folder path> For example, `C:\temp`
Linux	/<folder path>/ For example, `/root/path`
Windows shared location	<folder path> with additional slashes (\) For example, the path `\\INV12B2B01\Shared\path`, is specified as `\\\\INV12B2B01\\Shared\\path`

Network security

MassIngestionRuntime uses internal service ports. These ports bind to the local host (127.0.0.1), so only processes running on the same machine as the Secure Agent can connect to them.

CMI Streaming Agent

Use the CMI Streaming Agent to define and deploy streaming ingestion and replication tasks. You configure streaming ingestion and replication tasks in the Data Ingestion and Replication service.

A CMI Streaming Agent runs on an on-premise system and works in conjunction with the Streaming Ingestion and Replication. In an on-premise system, the CMI Streaming Agent runs the jobs deployed by Streaming Ingestion and Replication. The agent provides status and statistics updates of each job.

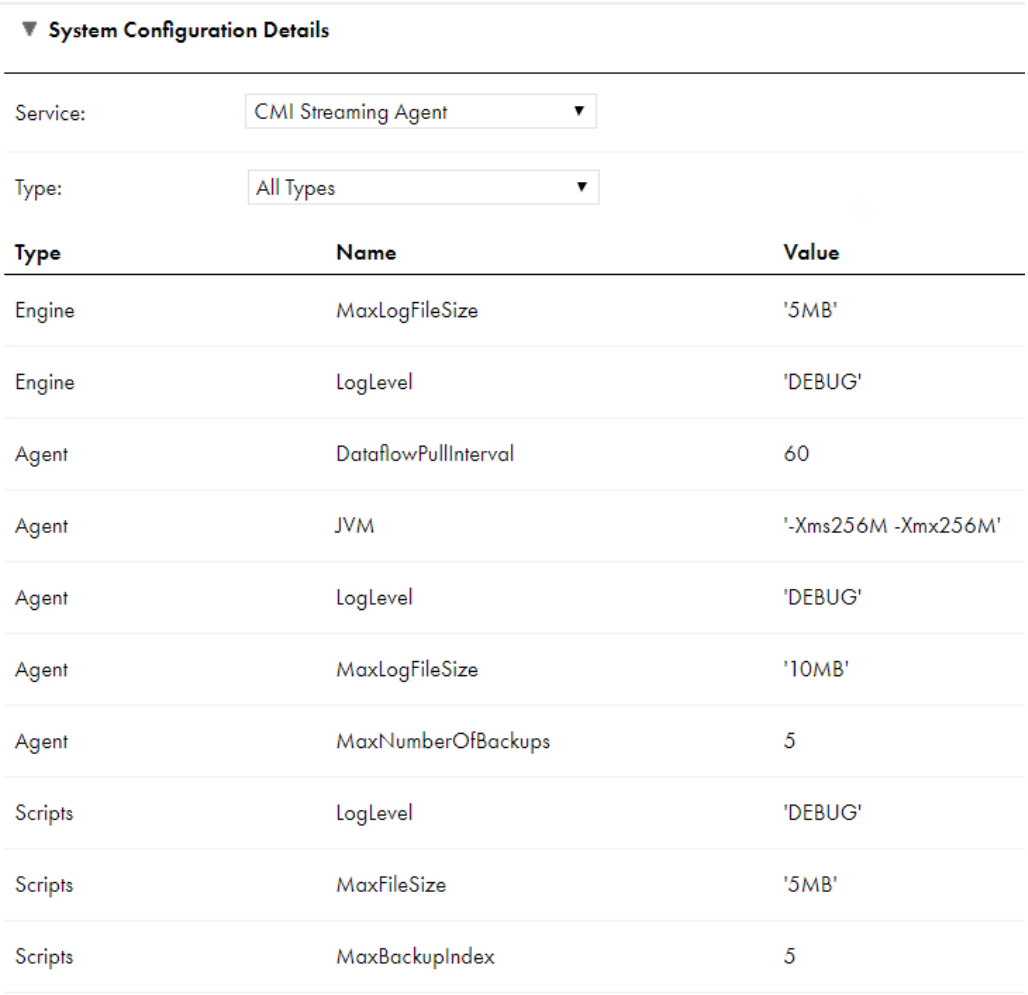
On Linux, the CMI Streaming Agent does not start if the agent installation directory name contains a space. The agent returns a connection timeout status. After a few restart attempts, the agent goes into the error state.

CMI Streaming Agent properties

To change or optimize the behavior of the CMI Streaming Agent, configure agent properties for your run-time environment. Configure CMI Streaming Agent properties in the **System Configuration Details** area when you edit a Secure Agent.

You can configure Engine, Agent, and Script properties of a CMI Streaming Agent.

The following image shows some of the CMI Streaming Agent properties:



System Configuration Details

Service: CMI Streaming Agent

Type: All Types

Type	Name	Value
Engine	MaxLogFileSize	'5MB'
Engine	LogLevel	'DEBUG'
Agent	DataflowPullInterval	60
Agent	JVM	'-Xms256M -Xmx256M'
Agent	LogLevel	'DEBUG'
Agent	MaxLogFileSize	'10MB'
Agent	MaxNumberOfBackups	5
Scripts	LogLevel	'DEBUG'
Scripts	MaxFileSize	'5MB'
Scripts	MaxBackupIndex	5

You can configure the following CMI Streaming Agent properties:

Type	Property Name	Description
Engine	MaxLogFileSize	The maximum size of the log file that the engine can create. Default is 5 MB.
Engine	LogLevel	The log level for the engine.
Agent	DataflowPullInterval	The time interval after which the agent checks for updates in the task. Default is 60 seconds.
Agent	JVM	List of JVM properties for the agent. For example: `[-Xms256M -Xmx256M]`
Agent	LogLevel	The log level for the agent.
Agent	MaxLogFileSize	Maximum size of the log files that an agent can create. Default is 10 MB.

Type	Property Name	Description
Agent	MaxNumberOfBackups	Maximum number of backup log files for the agent. Default is 5.
Scripts	LogLevel	The log level of the scripts.
Scripts	MaxFileSize	The maximum file size after which the log rolls over and creates a new file. Default is 10 MB.
Scripts	MaxBackupIndex	Maximum number of backup files maintained after rolling over. Default is 5.

Streaming Agent offline mode

You can run and monitor a streaming ingestion and replication job when the CMI Streaming Agent is offline or not connected to the internet.

The Streaming Agent supports both online and offline modes of communication. In the offline mode, the streaming ingestion and replication job continues to run even if the Streaming Agent does not communicate with the IDMC for an extended period of time. The Streaming Agent continues to monitor the health and statistics of the jobs locally. When the Streaming Agent turns online and connects to the cloud services, it updates any configuration changes for the agent and jobs, as well as updates the health and statistics to the services.

To switch between the offline and online modes, you can use the command line utility provided by the Streaming Ingestion and Replication service. Run the following command to start the command line utility:

```
<Informatica Secure Agent>/apps/Streaming_Ingestion_Agent/<version>/runcli.bat
```

The command line utility uses the command prompt `infa/stream>` and provides three groups of commands.

You can change the communication modes only through the command line utility. The Streaming Agent preserves the communication mode when the agent restarts.

The following table lists the commands of this command line utility:

Command	Description	Example

app-config	Shows the current configuration of the Streaming Agent application.	`infa/stream :>app-config`
		`deploy.pull.interval : 60`
		`health.poll.interval : 30`
		`minifi.ingester.file.location : ./conf`
		`siagent.communication.mode : Online`
		`siagent.monitoring.persist.dir : ../data`
		`siagent.statistics.post.batchsize : 720`
		`siagent.statistics.post.concurrency : 60`
		`siagent.status.persist.dir : ../data`
		`statistics.poll.interval : 30`

app-setconfig	Use this command to configure the following properties:
	- `siagent.communication.mode`. Use to configure offline or online communication mode.
	- `siagent.statistics.post.batchsize`. Use to define the number of snapshots in a batch.
	- `siagent.statistics.post.concurrency`. Use to define the number of worker threads to post statistics.
	The `--key` and `--value` tokens are optional.
		`infa/stream :>app-setconfig --key siagent.statistics.post.batchsize --value 20`
		or
		`infa/stream :>app-setconfig siagent.statistics.post.batchsize 20`

app-status	Shows the current status of the Streaming Agent.
	The `health status code` and `health error message` indicates the status of the agent (service) shown on the **Administrator**.
	`uptime` indicates the number of seconds since the Streaming Agent application is available.
		`infa/stream :>app-status`
		`health error message : No errors`
		`health status code : RUNNING(0)`
		`uptime : 67828`

app-statistics	Shows metadata and status of overall statistics collection in the Streaming Agent.
	- `collection interval`. Interval of statistics collection, in seconds.
	- `post interval`. Frequency of statistics posted or attempted post.
	- `max batch size`. Maximum number of snapshots posted in a single http post.
	- `last batch size`. Number of snapshots in the last http post.
	- `last time collected`. Timestamp when any statistics were last collected.
	- `last time posted`. Timestamp when any statistics were last posted.
		`infa/stream :>app-statistics`
		`collection interval : 30`
		`last batch size : 2`
		`last time collected : 7/3/20 10:19:03 AM IST`
		`last time posted : 7/3/20 10:18:53 AM IST`
		`max batch size : 20`
		`pending snapshots : 3`
		`post interval : 30`

Command	Description	Example
clear	Clears the screen.	-
exit, quit	Quits the application.	-
help	Shows a summary of all the commands available.	infa/stream :>help AVAILABLE COMMANDSAgent Application Commands app-config: Show agent application configuration app-setmode: Set the communication mode [Online/Offline] app-status: Show agent application statusBuilt-In Commands clear: Clear the shell screen. exit, quit: Exit the shell. help: Display help about available commands.Streaming Ingestion Task Commands task-health: Show streaming ingestion task health task-list: Show streaming ingestion task list task-metadata: Show streaming ingestion task metadata
task-list	Shows the list of streaming ingestion and replication jobs currently deployed on the Streaming Agent.	infa/stream :>task-list 6e61e76f-2618-4292-ab3d-dd181f47ee91 ad5053c7-5ac2-493f-8cbb-a24900b61f71
task-health	Shows the health status of all streaming ingestion and replication jobs in the Streaming Agent. Use the options `--name` or `--id` to specify a job. If none are specified, all jobs are listed.	infa/stream :>task-health --name aby_df4 processors : [{"id":"14a7a095-7fac-4fc3-ac5c-705369132516","status":"ERROR"},{"id":"821e6730-3aed-4d3f-b875-45f424b6b963","status":"RUNNING"}] status : ERROR timestamp : Sat May 09 06:04:08 IST 2020 infa/stream :>task-health 6e61e76f-2618-4292-ab3d-dd181f47ee91 processors : [{"id":"2a0b8715-aa7a-46c5-9d6a-6a356f5a0102","status":"ERROR"},{"id":"1172f3a8-35dd-41ef-be4b-bc0cf37e3794","status":"RUNNING"}] status : ERROR timestamp : Sat May 09 06:04:08 IST 2020 ad5053c7-5ac2-493f-8cbb-a24900b61f71 processors : [{"id":"14a7a095-7fac-4fc3-ac5c-705369132516","status":"ERROR"},{"id":"821e6730-3aed-4d3f-b875-45f424b6b963","status":"RUNNING"}] status : ERROR timestamp : Sat May 09 06:04:08 IST 2020

Command	Description	Example

task-metadata	Shows the metadata of all streaming ingestion and replication jobs in the Streaming Agent.
	Use the options `--name` or `--id` to specify a job.
	If none are specified, all jobs are listed.
		`infa/stream :>task-metadata --name aby_df4`
		`id : ad5053c7-5ac2-493f-8cbb-a24900b61f71`
		`name : aby_df4`
		`runId : 9071 version : 1`
		`infa/stream :>task-metadata`
		`6e61e76f-2618-4292-ab3d-dd181f47ee91`
		`id : 6e61e76f-2618-4292-ab3d-dd181f47ee91`
		`name : aby_df2`
		`runId : 9069`
		`version : 8`
		`ad5053c7-5ac2-493f-8cbb-a24900b61f71`
		`id : ad5053c7-5ac2-493f-8cbb-a24900b61f71`
		`name : aby_df4`
		`runId : 9071`
		`version : 1`

task-statistics	Shows the statistics details of all streaming ingestion and replication jobs in the Streaming Agent.
	Use the options `--name` or `--id` to specify a job.
	If none are specified, all jobs are listed.
		`infa/stream :>task-statistics --name aby_df1`
		`dataflow name : aby_df1`
		`last time collected : 1590861803731`
		`last time posted : 1590861806091`
		`infa/stream :>task-statistics`
		`7b7d3c09-df43-482f-b6c8-8dd80187e6d7`
		`dataflow name : aby_df2`
		`last time collected : 1590861770731`
		`last time posted : 1590861741132`
		`decfad0a-20df-4226-84f9-1ff1ab6ef96a`
		`dataflow name : aby_df1`
		`last time collected : 1590861768730`
		`last time posted : 1590861771054`

Online mode to Offline mode

By default, the Streaming Agent is in online mode.

To change the Streaming Agent to offline mode:

1. Launch the command line utility using the following command:
   In Windows:

   ```
   <Informatica_Secure_Agent>/apps/Streaming_Ingestion_Agent/<version>/runcli.bat
   ```

   In UNIX:

   ```
   <Informatica_Secure_Agent>/apps/Streaming_Ingestion_Agent/<version>/runcli.sh
   ```

2. Set the Streaming Agent to offline mode:

   ```
   app-setconfig --key siagent.communication.mode --value Offline
   ```

or

```
app-setconfig siagent.communication.mode Offline
```

The Streaming Agent stops sending health updates and statistics of any streaming ingestion and replication job.

Offline mode to Online mode

To change the Streaming Agent to online mode:

1. Launch the command line utility using the following command:
   In Windows:

   ```
   <Informatica_Secure_Agent>/apps/Streaming_Ingestion_Agent/<version>/runcli.bat
   ```

   In UNIX:

   ```
   <Informatica_Secure_Agent>/apps/Streaming_Ingestion_Agent/<version>/runcli.sh
   ```

2. Set the Streaming Agent to online mode:

   ```
   app-setconfig --key siagent.communication.mode --value Online
   ```

   or

   ```
   app-setconfig siagent.communication.mode Online
   ```

   The Streaming Agent starts sending health updates of all the streaming ingestion and replication jobs and the updates appear in the **Monitoring** page. It starts sending statistics of all the streaming ingestion and replication jobs including the statistics backlog collected while it was offline to the service. It also synchronizes the updates to the streaming ingestion and replication jobs or adds the new streaming ingestion and replication job deployed while it was offline to the service.

Agent failover in a Secure Agent group

If the active agent in a Secure Agent group goes down unexpectedly while an application ingestion and replication job or a database ingestion and replication job is running, the job can fail over to another available agent in the group.

Application Ingestion and Replication and Database Ingestion and Replication support automatic failovers for initial load jobs that have any supported source type. Database Ingestion and Replication also supports automatic failovers for incremental load and combined initial and incremental load jobs that have a Db2 for i, Db2 for LUW, Db2 for z/OS, MongoDB, MySQL, Oracle, PostgreSQL, SAP HANA, or SQL Server source, subject to the following limitations:

- Jobs cannot have persistent storage enabled.
- Jobs that have Kafka targets must store checkpoint information in the Kafka header. For any jobs that existed before the July 2025 release, automatic switchovers can't occur because checkpoint information is stored in the checkpoint file in the Secure Agent.
- Jobs that use the Query-based CDC method to capture changes from a Db2 for LUW, Oracle, or SQL Server source must have an Amazon Redshift, Amazon S3, Databricks, Google BigQuery, Google Cloud Storage, Microsoft Azure Data Lake Storage, Microsoft Azure Synapse Analytics, Microsoft Fabric OneLake, Oracle, Oracle Cloud Object Storage, PostgreSQL, Snowflake, or SQL Server target.
- Jobs that have an SAP HANA source cannot use the trigger-based CDC method.
- If you use a read-only replica for a PostgreSQL connection, you must set the readerPostgreSQLShouldDisableLocalPersistent custom property to false on the **Task Details** source page of the task configuration wizard .

Application Ingestion and Replication also supports automatic failovers for incremental load and combined initial and incremental load jobs across all supported source and target types, except for SAP ODP sources and Kafka targets, subject to the following limitations:

- Jobs cannot have persistent storage enabled.
- If the Database Ingestion Service is in a stopped state, you need to manually stop and then resume the job.

Automatic failover occurs after the 15-minute heartbeat interval of the original agent expires. Manual failover can be triggered by aborting the job and then resuming it while the original agent is down.

Failover can be affected by how you set the supportedLoadTypes property of Database Ingestion agent service for the agents in the group. You can set the property separately for each agent in the group such that different agents run different load types. In this scenario, failover can occur only if another agent in the group is configured for the same load type as the failed agent or is configured for both load types (Initial and Incremental). For example, if you set the supportedLoadTypes property to Incremental for agent A, Initial for agent B, and Initial and Incremental for agent C, when agent A goes down, the job cannot fail over to agent B but it can fail over to agent C. Alternatively, if agent C goes down, the job might fail over to either agent A or Br, depending on internal load-type matching.

Also, failover is affected by agent capacity. If agent A fails, but agent C is already running with a full load, no failover occurs until capacity on agent C is available. For example, if a job that's running on agent C stops, enough capacity might become available for the failover from agent A to agent C to occur.

Another consideration is that all agents in a Secure Agent group are used for load balancing as well as failover. If multiple agents support the same load, jobs of that load type can be distributed across them. However, you can't specify which agent would be used for failover.

Serverless runtime environments

A serverless runtime environment is an advanced serverless deployment solution that doesn't require downloading, installing, configuring, or maintaining a Secure Agent or Secure Agent group. You can use a serverless runtime environment in the same way that you use a runtime environment when you configure a connection or tasks.

A serverless runtime environment is an advanced serverless deployment solution that uses an isolated, single-tenant model.

The single-tenant model provides a dedicated server with virtual machine resources to run tasks for your organization. The serverless runtime environment auto-scales with the size of the workload while your data remains in your cloud environment.

For Data Ingestion and Replication, a serverless runtime environment can be hosted on the Azure Virtual Network (VNet) cloud platform.

For more information about serverless runtime environments, see *Administrator* > *Runtime environments* > *Serverless runtime environments*.

You can run the following types of application ingestion and replication jobs and database ingestion and replication jobs on a serverless runtime environment:

Application Ingestion and Replication sources:

- For all load types:

  - Marketo

  - Microsoft Dynamics 365
  - NetSuite
  - Oracle Fusion Cloud
  - Salesforce
  - SAP ECC on Oracle database using the SAP Mass Ingestion connector
  - SAP OData V2
  - ServiceNow
  - Workday
- For initial loads:
  - Salesforce Marketing Cloud

These application ingestion and replication jobs can have any of the following target types: Amazon Redshift, Amazon S3, Databricks, Google BigQuery, Google Cloud Storage, Microsoft Azure Data Lake Storage Gen2, Microsoft Azure Synapse Analytics, Microsoft Fabric OneLake, Oracle, Oracle Cloud Object Storage, PostgreSQL, Snowflake without the Superpipe option, and SQL Server.

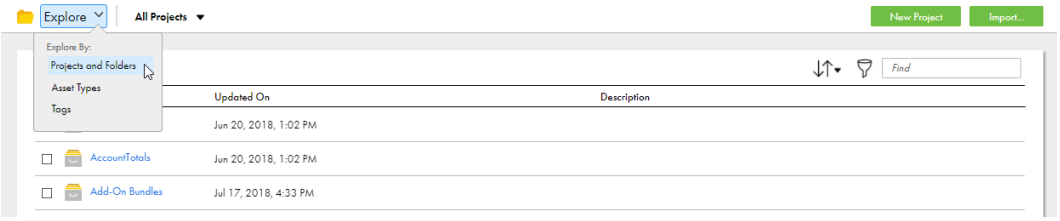
Database Ingestion and Replication sources:

- For all load types with any target type:
  - Db2 for i
  - Db2 for z/OS
  - MongoDB
  - MySQL
  - Oracle (log-based CDC method only)
  - PostgreSQL (except for PostgreSQL read-only replicas)
  - SQL Server (CDC tables and log-based CDC methods only)
- For initial loads with any target type:
  - SAP HANA
- For incremental loads and combined initial and incremental loads with Amazon Redshift, Amazon S3, Databricks, Google BigQuery, Google Cloud Storage, Microsoft Azure Data Lake Storage Gen2, Microsoft Azure Synapse Analytics, Microsoft Fabric OneLake, Oracle, Oracle Cloud Object Storage, PostgreSQL, Snowflake without the Superpipe option, or SQL Server:
  - Oracle (query-based CDC method only)
  - SQL Server (query-based CDC method only)

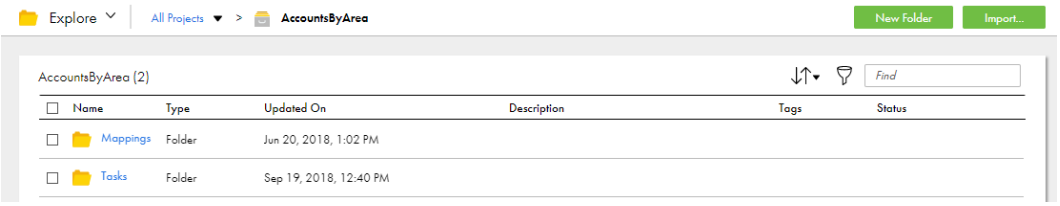
Creating projects and project folders

Projects can contain multiple folders that you can use to organize the assets used in the project. Create projects using the **Explore** page.

To create a project, go to the **Explore** page and select to explore by projects and folders, and then click **New Project**.



To create a project folder, go to the **Explore** page and open the project, and then click **New Folder**.



You can create one level of folders in a project. You cannot create folders within folders.

For more information about working with projects, see *Asset Management*.

Editing your user profile

Your user profile contains the details of your IDMC user account.

You can update the following information in your profile:

- First and last name
- Job title
- Email address
- Phone number
- Time zone (used in the job execution time stamps on the **All Jobs**, **Running Jobs**, **My Jobs**, **Import/Export Logs**, and **My Import/Export Logs** pages)
- Password
- Security question and answer

Note: If you use SAML to log in to IDMC and your organization administrator has enabled SAML group and role mapping on the **SAML Setup** page in Administrator, you can update the time zone attribute. Your enterprise directory updates the other attributes directly each time you log in to IDMC.

1. Click the **User** icon in the top right corner of the IDMC window and then select **Profile**.
2. On the **Profile** page, add or edit personal information such as your name, job title, phone number, and time zone.

3. To update your email address, click **Update Email**.

   After you enter your password to initiate the change, IDMC sends a verification email to your new email address. The email contains a link that's valid for 24 hours. When you click the link in the email, IDMC verifies the new address, and it appears in your profile. If the link expires, you can resend the verification email.
   If multi-factor authentication is enabled in your organizaion, IDMC sends a verification code to your original email address. You must enter the code in the dialog box on your **Profile** page to receive the verification link in your new email address.
4. Optionally, change your password or security question.
5. Click **Save**.

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